Math 252 (23526) **Quiz 6** **Name:** ____________________

1. Use trigonometric substitution to find $\displaystyle\int \frac{1}{x^2\sqrt{x^2-4}}\,dx$.

(Note: You will not receive credit if you use other methods.)

$x = 2\sec\theta$

$dx = 2\sec\theta\tan\theta\,d\theta$

$\sqrt{x^2-4} = \sqrt{4\sec^2\theta-4} = \sqrt{4(\sec^2\theta-1)} = \sqrt{4\tan^2\theta} = 2\tan\theta$

$$\int \frac{1}{x^2\sqrt{x^2-4}}\,dx = \int \frac{1}{4\sec^2\theta\cdot 2\tan\theta}\cdot 2\tan\theta\cdot\sec\theta\,d\theta$$

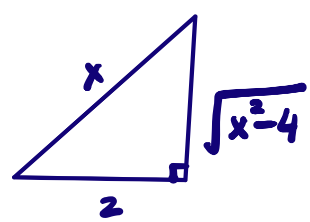
$$= \frac{1}{4}\int \frac{1}{\sec\theta}\,d\theta$$

$$= \frac{1}{4}\int \cos\theta\,d\theta$$

$$= \frac{1}{4}\sin\theta + C$$

$$= \frac{1}{4}\cdot\frac{\sqrt{x^2-4}}{x} + C$$

$\frac{x}{2} = \sec\theta$



Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1-\cos 2x}{2}, \quad \cos^2 x = \frac{1+\cos 2x}{2}, \quad \sin 2x = 2\sin x\cos x$$

$$\int \cot x\csc x\,dx = -\csc x + C, \quad \int \csc^2 x\,dx = -\cot x + C$$

$$\int \tan x\,dx = \ln|\sec x| + C, \quad \int \cot x\,dx = \ln|\sin x| + C$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C, \quad \int \csc x\,dx = \ln|\csc x - \cot x| + C$$